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The Effect of Minimum Wages on Low-Wage Jobs

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1 Big picture

1.1 Research question

What is the causal effect of minimum-wage increases on employment, especially at the bottom of the wage distribution?

1.2 Why standard approaches struggle (the core identification issue)

Conventional designs often estimate an aggregate employment elasticity:

$$\Delta \ln(E_{st}) = \beta \Delta \ln(\text{MW}_{st}) + \text{controls} + \varepsilon_{st},$$

but in U.S. data, only a small fraction of workers is directly affected by the minimum wage. This creates two empirical challenges:

1. **Low signal-to-noise:** aggregate employment includes many “upper-tail” jobs that the policy should not affect. Noise and confounders in these upper-tail jobs can dominate.
2. **Weak “first stage”:** if minimum wages barely move average wages for broad groups, interpreting any estimated employment change is difficult.

1.3 Core idea: locate the effect inside the wage distribution

Rather than starting with aggregate employment, the paper starts with the **wage distribution** and asks:

How many jobs disappear below the new minimum wage, and where do those jobs go?

This shifts the empirical focus to the part of the outcome space where the policy should matter.

2 Conceptual framework: missing jobs, excess jobs, and the key accounting identity

2.1 Frequency distribution logic (Figure I)

Let $\text{Emp}[w \in B]$ denote employment in a wage interval (bin) B . A binding minimum wage at MW should move jobs that used to pay $w < \text{MW}$. Define:

- **Missing jobs below the new minimum wage:** jobs that (counterfactually) would have paid below MW but no longer do.
- **Excess jobs at/above the new minimum wage (up to some upper bound W):** jobs that now pay at least MW because wages were raised.

In the figure-based conceptual decomposition:

$$\Delta b \equiv Emp_1[w < MW] - Emp_0[w < MW] \quad (\text{missing jobs below MW}), \quad (1)$$

$$\Delta a \equiv Emp_1[MW \leq w < W] - Emp_0[MW \leq w < W] \quad (\text{excess jobs above MW}), \quad (2)$$

$$\Delta e \equiv \Delta a + \Delta b \quad (\text{net employment effect in the affected range}). \quad (3)$$

Here $Emp_1[\cdot]$ is the observed post-policy distribution and $Emp_0[\cdot]$ is the counterfactual distribution absent the policy.

2.2 Interpretation

- If minimum wages *only raise wages without destroying jobs*, then $\Delta b < 0$ and $\Delta a \approx -\Delta b$ so that $\Delta e \approx 0$.
- If minimum wages destroy jobs, then $|\Delta b|$ exceeds the offsetting increase above MW, so $\Delta e < 0$.

This simple accounting is the spine of the paper: the empirical work estimates Δa and Δb using an event-study design.

3 Data and sample construction

- **Wage and individual employment:** CPS MORG (1979–2016). Hourly wages for hourly workers; weekly earnings / weekly hours for others.
- **Benchmarking:** employment levels are benchmarked to QCEW (near universe of employment), which mainly improves precision.
- **Events:** 138 eligible state minimum-wage increases (main sample). Federal and very small changes are handled via controls.

4 All models in the paper (with interpretation)

4.1 Model 1: main event-study-by-wage-bin regression (Equation (1))

The main estimating equation is a **stacked event-study DD** run on state $\times$ quarter $\times$ wage-bin cells:

$$\frac{E_{sjt}}{N_{st}} = \sum_{\tau=-3}^4 \sum_{k=-4}^{17} \alpha_{\tau k} I_{sjt}^{\tau k} + \mu_{sj} + \rho_{jt} + \Gamma_{sjt} + u_{sjt}. \quad (4)$$

Objects:

- E_{sjt} : employment in state s , quarter t , in a \$0.25 wage bin indexed by j .
- N_{st} : population in state s and quarter t (so E_{sjt}/N_{st} is per-capita employment in that wage bin).

- Event time τ : years relative to the minimum-wage increase (annualized; each “year” is 4 quarters). $\tau = 0$ is the first year after the hike; $\tau = -1$ is the year before.
- Relative-wage bin k : bins defined *relative to the new minimum wage*. Example: $k = 0$ is wages in $[\text{MW}, \text{MW} + 0.99]$ and $k = -1$ is wages in $[\text{MW} - 1.00, \text{MW} - 0.01]$.
- $I_{s jt}^{\tau k}$: indicator that the observation (s, j, t) belongs to event time τ and relative-wage dollar bin k for the relevant event.

Fixed effects / controls:

- μ_{sj} : state-by-wage-bin fixed effects (absorbs permanent state-specific wage distributions).
- ρ_{jt} : period-by-wage-bin fixed effects (absorbs national time-varying changes in wage inequality).
- $\Gamma_{s jt}$: controls for federal and very small minimum-wage changes so they do not contaminate the main events.

Standard errors are clustered at the state level.

4.2 From coefficients to “missing” and “excess” jobs (Equation (1) $\Rightarrow$ objects)

Choose an upper bound W for potential spillovers. The baseline uses $W = \text{MW} + 4$ (i.e., $k = 0, 1, 2, 3, 4$). Using the estimated coefficients, define for each event time τ :

Missing jobs below the new minimum wage.

$$\Delta b_{\tau} \equiv \sum_{k=-4}^{-1} (\alpha_{\tau k} - \alpha_{-1, k}). \quad (5)$$

Excess jobs at/above the new minimum wage (up to W).

$$\Delta a_{\tau} \equiv \sum_{k=0}^4 (\alpha_{\tau k} - \alpha_{-1, k}). \quad (6)$$

Net employment change in the affected range.

$$\Delta e_{\tau} \equiv \Delta a_{\tau} + \Delta b_{\tau}. \quad (7)$$

Normalization used in figures/tables. Let EPOP_{-1} be the average employment-to-population ratio in treated states in the year before treatment. Then:

$$\widehat{\Delta a_{\tau}} / \text{EPOP}_{-1}, \quad \widehat{\Delta b_{\tau}} / \text{EPOP}_{-1}$$

are “excess” and “missing” jobs expressed as a fraction of pre-treatment total employment.

Five-year averaged effects. The paper often reports averages over the post window $\tau = 0, \dots, 4$:

$$\Delta a \equiv \frac{1}{5} \sum_{\tau=0}^4 \Delta a_{\tau}, \quad \Delta b \equiv \frac{1}{5} \sum_{\tau=0}^4 \Delta b_{\tau}, \quad \Delta e \equiv \Delta a + \Delta b.$$

4.3 Policy-relevant elasticities constructed from the decomposition

Let b_{-1} be the pre-treatment fraction of jobs below the new minimum wage.

Percent change in affected employment.

$$\% \Delta \text{Affected Employment} \equiv \% \Delta e = \frac{\Delta a + \Delta b}{b_{-1}}. \quad (8)$$

Employment elasticity w.r.t. the minimum wage. Let $\% \Delta \text{MW}$ be the average proportional change in the minimum wage across events.

$$\varepsilon_{E, \text{MW}} \equiv \frac{\% \Delta \text{Total Employment}}{\% \Delta \text{MW}} = \frac{\Delta a + \Delta b}{\% \Delta \text{MW}}. \quad (9)$$

4.4 Model 2: wage effects and the wage-bill identity (Equation (2))

Let wb_{-1} denote the pre-treatment wage bill earned by workers below the new minimum wage and let $w_{-1} = wb_{-1}/b_{-1}$ be their average wage. Let Δwb be the policy-induced change in the wage bill for affected workers and Δe the change in affected employment. Then the post-policy average wage for affected workers is:

$$w = \frac{wb_{-1} + \Delta wb}{b_{-1} + \Delta e}.$$

So the percent change in the affected wage is:

$$\% \Delta w = \frac{w}{w_{-1}} - 1 = \frac{wb_{-1} + \Delta wb}{b_{-1} + \Delta e} \cdot \frac{b_{-1}}{wb_{-1}} - 1 = \frac{\% \Delta wb - \% \Delta e}{1 + \% \Delta e}. \quad (10)$$

This formula makes clear that wage effects depend on both the wage-bill increase and any employment change among affected workers.

Own-wage (labor demand) elasticity for affected workers.

$$\varepsilon_{E, w} \equiv \frac{\% \Delta \text{Affected Employment}}{\% \Delta \text{Affected Wage}} = \frac{1}{\% \Delta w} \cdot \frac{\Delta a + \Delta b}{b_{-1}}. \quad (11)$$

4.5 Model 3: “no-spillover” wage effect (Equation (3)) and spillover share

The total wage change $\% \Delta w$ includes direct effects (bringing sub-minimum jobs up to MW) and spillovers above MW. The paper defines a “no-spillover” wage change by moving missing jobs exactly to MW:

$$\% \Delta w^{\text{no spillover}} = \frac{\sum_{k=-4}^{-1} k (\alpha_k - \alpha_{-1, k})}{wb_{-1}}, \quad (12)$$

where α_k denotes the post-window average coefficient $\alpha_k \equiv \frac{1}{5} \sum_{\tau=0}^4 \alpha_{\tau k}$.

Then the implied **spillover share** of the affected-wage increase is:

$$\text{Spillover share} \equiv \frac{\% \Delta w - \% \Delta w^{\text{no spillover}}}{\% \Delta w}. \quad (13)$$

4.6 Model 4: event-by-event estimation for heterogeneity (Equation (4))

To study heterogeneity across specific minimum-wage events h , the paper constructs an eight-year panel around each event with “clean” control states (no other nontrivial changes in that window) and estimates:

$$Y_{sjt}^{(h)} = \sum_{\tau=-3}^4 \sum_{k=-4}^4 \alpha_{\tau k}^{(h)} I_{sjt}^{\tau k, (h)} + \mu_{sj}^{(h)} + \rho_{jt}^{(h)} + \Gamma_{sjt}^{(h)} + u_{sjt}^{(h)}. \quad (14)$$

Here j indexes \$1 bins relative to the new minimum wage (so $k = -4, \dots, 4$), and the resulting event-specific $\Delta a_h, \Delta b_h, \Delta e_h$ are computed analogously to the pooled analysis. These event-specific estimates are then related to bite measures such as the Kaitz index (Figure V).

4.7 Model 5: TWFE on log minimum wages (diagnostic comparison; Figure VI)

Although not the paper’s preferred model, the authors also run standard two-way fixed effects regressions with leads/lags of $\ln \text{MW}_{st}$, separately for each wage bin, and cumulate post-treatment responses. Figure VI shows that these TWFE-logMW models can produce sizable spurious employment movements in upper-tail wage bins, and those movements can drive misleading aggregate elasticities.

5 Interpretation of all figures (I–VI)

5.1 Figure I: the conceptual decomposition in the wage distribution

Figure I illustrates the identification target: minimum wages create (i) missing jobs below the new minimum and (ii) excess jobs at and just above it. The *net* employment effect in the affected range is $\Delta e = \Delta a + \Delta b$. The key empirical question is whether Δa offsets Δb .

5.2 Figure II: main empirical histogram (employment changes by \$1 bin)

Figure II plots five-year averaged post-treatment employment changes in each \$1 bin relative to the new minimum wage. The figure shows:

- a sharp **decline in jobs below the new minimum wage** (large negative bars for $k < 0$),
- a large **spike at the minimum wage** (positive bar at $k = 0$),
- modest **spillovers** into bins $k = 1, 2, 3, 4$,
- **no meaningful changes in the upper tail** ($k \geq 5$), and therefore a running sum near zero at high wages.

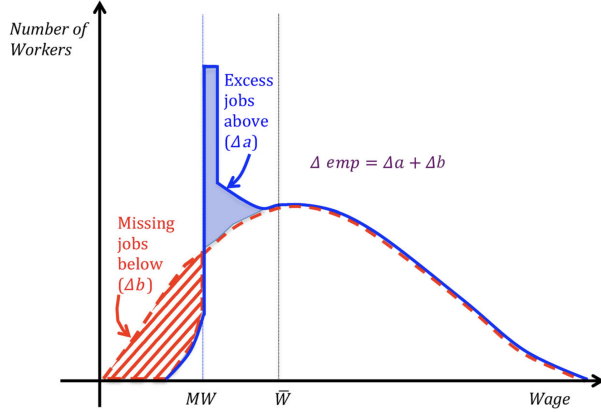


FIGURE I
The Impact of Minimum Wages on the Frequency Distribution of Wages

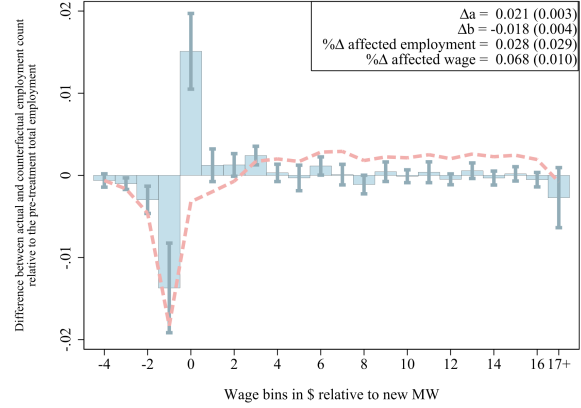


FIGURE II
Impact of Minimum Wages on the Wage Distribution

The inset summarizes the accounting: missing jobs $\Delta b \approx -0.018$ and excess jobs $\Delta a \approx +0.021$ (fractions of pre-treatment employment), so the net effect is close to zero, while affected wages rise substantially.

5.3 Figure III: dynamics of missing vs excess jobs over event time

Figure III plots Δb_τ and Δa_τ over event time $\tau = -3, \dots, 4$. Key messages:

- minimal pre-trends in the years leading up to the hike (leads near zero),
- an immediate negative jump in missing jobs (Δb_τ) at $\tau = 0$,
- an immediate positive jump in excess jobs (Δa_τ) of similar magnitude,
- persistence of both effects for several years, indicating durable treatment.

This figure supports the interpretation that minimum wages mainly reallocate jobs from below-MW wages to at/above-MW wages.

5.4 Figure IV: entrants vs incumbents (one-year horizon)

Figure IV separates **new entrants** (not employed one year before the hike) from **incumbents** (employed one year before). The figure shows that the **ripple/spillover effect** above the minimum wage is primarily driven by incumbents: incumbent workers exhibit larger positive employment changes in bins above MW (conditional on remaining employed), whereas entrants show less of a spillover pattern. This helps interpret wage spillovers as largely reflecting within-job wage adjustments for continuing matches.

5.5 Figure V: heterogeneity across events and bite (Kaitz index)

Figure V uses event-by-event estimates and relates them to the Kaitz index (new minimum wage divided by the median wage at the time of the hike). The plots show:

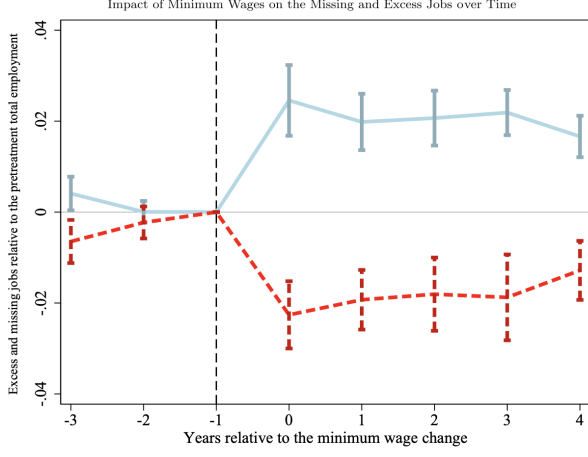


FIGURE III

Impact of Minimum Wages on the Missing and Excess Jobs over Time

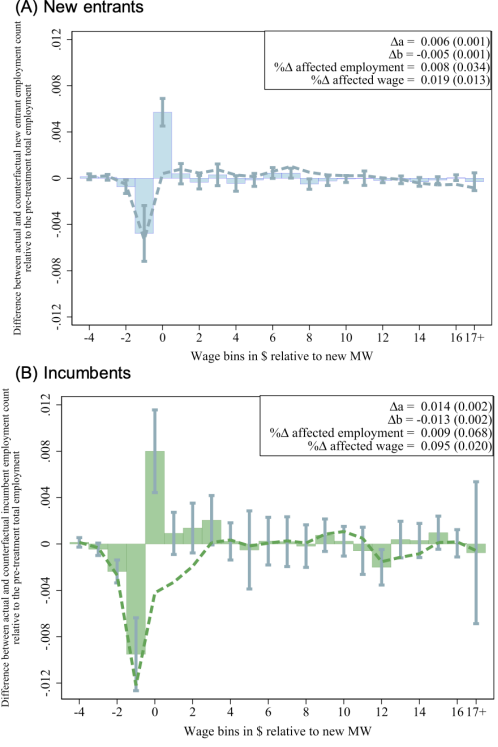


FIGURE IV

Impact of Minimum Wages on the Wage Distribution by Pretreatment Employment Status: New Entrants and Incumbents

- more binding hikes (higher Kaitz) are associated with more missing jobs (larger negative Δb_h in magnitude),
- but also more excess jobs (larger positive Δa_h),
- resulting in little systematic relationship between bite and *net* employment effects Δe_h .

This indicates that the “reallocation” pattern strengthens with bite but does not translate into sizeable job losses.

5.6 Figure VI: TWFE-logMW can be driven by upper-tail movements

Figure VI shows that conventional TWFE specifications on log minimum wages can produce nontrivial estimated employment changes in high-wage bins that are unlikely to be caused by minimum wages. When these upper-tail movements are summed, they can generate apparently large aggregate employment elasticities. The figure clarifies why focusing on the *local* wage region around MW is a more robust diagnostic and estimation strategy.

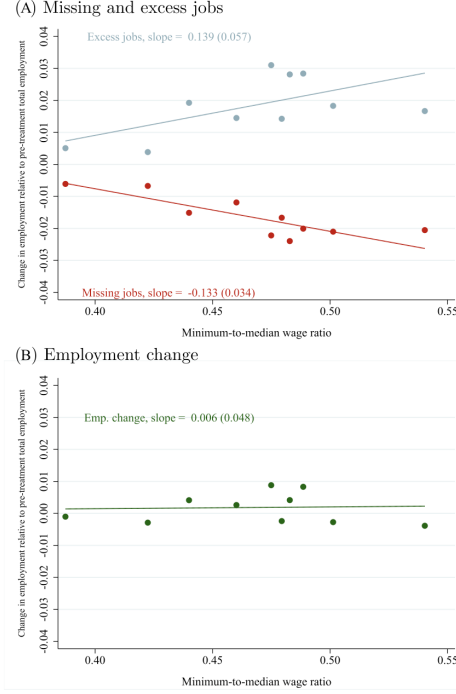


FIGURE V

Relationship between Excess Jobs, Missing Jobs, Employment Change, and the Minimum-to-Median Wage Ratio across Events

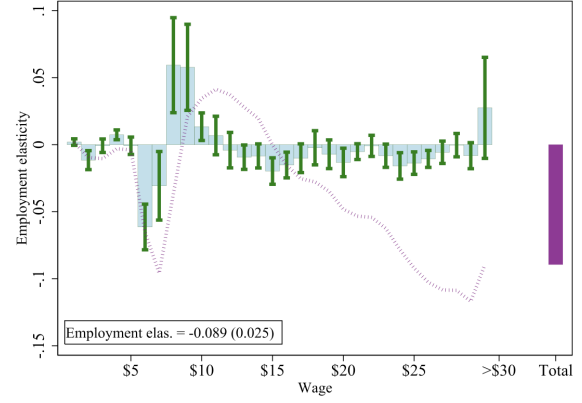


FIGURE VI

Impact on Employment throughout the Wage Distribution in the Two-Way Fixed Effects Model on Log Minimum Wages

6 Interpretation of all tables (I–IV)

6.1 Table I: baseline effects on employment and wages (main sample, 1979–2016)

What the table reports. Table I summarizes (five-year averaged) missing jobs (Δb), excess jobs (Δa), net employment effects in affected bins, affected-wage changes, and implied elasticities across alternative specifications (different fixed effects and trends).

Key empirical conclusions.

- **Missing jobs are large and robust:** Δb is about -0.015 to -0.018 of pre-treatment employment and always statistically significant.
- **Excess jobs largely offset:** Δa is about $+0.014$ to $+0.021$ and statistically significant.
- **Net employment effects are near zero:** $\% \Delta$ affected employment is small and statistically insignificant across columns.
- **Wage effects are sizable:** affected wages rise meaningfully (about 4.3% to 6.8% depending on specification).
- **Elasticities rule out large negatives:** implied employment elasticities w.r.t. the minimum wage and w.r.t. own wages are imprecise but centered near zero and inconsistent with large disemployment effects.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Missing jobs below new MW (Δb)	-0.018*** (0.004)	-0.018*** (0.004)	-0.018*** (0.004)	-0.016*** (0.002)	-0.016*** (0.002)	-0.015*** (0.002)	
Excess jobs above new MW (Δa)	0.021*** (0.003)	0.018*** (0.003)	0.020*** (0.003)	0.016*** (0.002)	0.014*** (0.003)	0.015*** (0.003)	
% Δ affected wages	0.068*** (0.010)	0.057*** (0.010)	0.068*** (0.012)	0.049*** (0.010)	0.043*** (0.010)	0.050*** (0.011)	0.065*** (0.010)
% Δ affected employment	0.028 (0.029)	0.000 (0.023)	0.022 (0.021)	-0.002 (0.021)	-0.019 (0.021)	-0.000 (0.023)	0.027 (0.028)
Employment elasticity w.r.t. MW	0.024 (0.025)	0.000 (0.020)	0.019 (0.018)	-0.001 (0.018)	-0.016 (0.018)	-0.000 (0.019)	0.023 (0.024)
Emp. elasticity w.r.t. affected wage	0.411 (0.430)	0.006 (0.402)	0.326 (0.313)	-0.032 (0.439)	-0.449 (0.574)	-0.003 (0.455)	0.410 (0.421)
Jobs below new MW (b_{-1})	0.086 (0.101)	0.086 (0.101)	0.086 (0.101)	0.086 (0.101)	0.086 (0.101)	0.086 (0.101)	0.086 (0.101)
% Δ MW	138 (138)	138 (138)	138 (138)	138 (138)	138 (138)	138 (138)	138 (138)
Number of events	847,314	847,314	847,314	847,314	847,314	847,314	14,484
Number of observations	4,694,104	4,694,104	4,694,104	4,694,104	4,694,104	4,694,104	4,694,104
Number of workers in the sample							
Controls							
Bin-state FE	Y	Y	Y	Y	Y	Y	Y
Bin-period FE	Y	Y	Y	Y	Y	Y	Y
Bin-state linear trends		Y	Y	Y	Y	Y	Y
Bin-state quadratic trends			Y				
Bin-division-period FE				Y	Y	Y	Y
State FE							Y
Year FE							Y

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Missing jobs below new MW (Δb)	-0.065*** (0.010)	-0.032*** (0.007)	-0.114*** (0.010)	-0.028*** (0.005)	-0.028*** (0.008)	-0.094*** (0.010)	-0.020*** (0.005)	-0.004*** (0.001)
Excess jobs above new MW (Δa)	0.075*** (0.011)	0.038*** (0.006)	0.127*** (0.020)	0.026*** (0.004)	0.028*** (0.006)	0.100*** (0.012)	0.021*** (0.003)	0.004*** (0.001)
% Δ affected wages	0.080*** (0.014)	0.076*** (0.014)	0.083*** (0.018)	0.072*** (0.011)	0.044*** (0.012)	0.073*** (0.011)	0.051*** (0.013)	0.060*** (0.032)
% Δ affected employment	0.038 (0.024)	0.043 (0.030)	0.030 (0.032)	0.025 (0.027)	-0.004 (0.044)	0.015 (0.018)	0.015 (0.048)	0.111 (0.055)
Employment elasticity w.r.t. MW	0.097 (0.061)	0.061 (0.042)	0.125 (0.134)	0.025 (0.027)	-0.005 (0.058)	0.052 (0.062)	0.016 (0.049)	0.003 (0.014)
Emp. elasticity w.r.t. affected wage	0.475* (0.268)	0.570 (0.386)	0.356 (0.317)	0.343 (0.362)	-0.086 (1.005)	0.206 (0.233)	0.304 (0.904)	0.184 (0.841)
Jobs below new MW (b_{-1})	0.264 (0.103)	0.145 (0.103)	0.432 (0.102)	0.102 (0.101)	0.133 (0.100)	0.358 (0.103)	0.104 (0.103)	0.027 (0.103)
% Δ MW								
Number of events	138	138	138	138	138	138	138	138
Number of observations	847,314	847,314	847,314	847,314	846,729	847,314	847,314	847,314
Number of workers in the sample	660,771	2,248,711	287,484	2,277,624	781,003	469,226	1,830,393	2,349,485
Sample	Less than high school	High school or less	Teen	Women	Black or Hispanic	High probability	Medium probability	Low probability

6.2 Table II: heterogeneity by demographic groups and exposure probability

Groups. Columns (1)–(5) examine groups with different baseline minimum-wage exposure (education, teens, women, Black/Hispanic). Columns (6)–(8) use demographic predictors (Card–Krueger style) to define high/medium/low probability of exposure.

Findings.

- Missing and excess jobs scale with exposure: the most exposed groups (e.g., teens; less education; high-probability group) show large $|\Delta b|$ and large Δa .
- Despite this, **net affected employment changes remain statistically indistinguishable from zero** for all groups, including the high-probability group where precision is highest.
- Affected wages rise for all groups (typically around 4–8%), consistent with wage compression at the bottom.
- The Black/Hispanic group shows smaller spillover-related patterns (connected to Table IV’s spillover shares), but still no robust job-loss pattern.

6.3 Table III: heterogeneity by sector (1992–2016)

Why start in 1992? Sector classification requires industry coding that is consistently available from 1992 onward.

Findings.

- The bite (b_{-1}) varies sharply: restaurants have very high exposure ($b_{-1} \approx 0.434$) whereas construction has very low exposure (≈ 0.036).
- Missing and excess jobs are largest where the minimum wage binds most (especially restaurants and nontradeables).
- Yet **net affected employment changes remain small and statistically insignificant** across sectors.
- Wage effects are clearly positive and large in sectors with strong bite (restaurants, nontradeables), consistent with the intended policy channel.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Missing jobs below new MW (Δb)	-0.019*** (0.004)	-0.016* (0.008)	-0.068*** (0.007)	-0.003 (0.002)	-0.011*** (0.003)	-0.101*** (0.015)	-0.023*** (0.003)	-0.017** (0.008)
Excess jobs above new MW (Δa)	0.020*** (0.003)	0.011 (0.008)	0.072*** (0.011)	0.005 (0.006)	0.011*** (0.002)	0.101*** (0.015)	0.041*** (0.010)	0.011 (0.009)
% Δ affected wages	0.058*** (0.011)	0.058 (0.073)	0.056*** (0.014)	0.097 (0.086)	0.056*** (0.013)	0.049*** (0.012)	0.060*** (0.021)	0.073 (0.078)
% Δ affected employment	0.008 (0.031)	-0.111 (0.136)	0.022 (0.037)	0.051 (0.163)	0.009 (0.044)	-0.001 (0.026)	0.062 (0.080)	-0.101 (0.145)
Employment elasticity w.r.t. MW	0.007 (0.027)	-0.056 (0.069)	0.060 (0.103)	0.019 (0.059)	0.005 (0.026)	-0.002 (0.117)	0.086 (0.111)	-0.052 (0.074)
Emp. elasticity w.r.t. affected wage	0.140 (0.523)	-1.910 (3.922)	0.387 (0.597)	0.530 (1.311)	0.166 (0.763)	-0.011 (0.542)	1.040 (1.058)	-1.385 (2.956)
Jobs below new MW (b_{-1})	0.087	0.050	0.270	0.036	0.057	0.434	0.136	0.050
% Δ MW	0.098	0.098	0.098	0.098	0.098	0.098	0.098	0.098
Number of events	118	118	118	118	118	118	118	118
Number of observations	554,931	554,931	554,931	554,931	554,931	554,931	554,931	554,931
Number of workers in the sample	2,652,792	358,086	384,498	274,812	1,504,643	156,634	315,397	349,749
Sector	Overall	Tradeable	Nontradeable	Construction	Other	Restaurants	Retail	Manufacturing

TABLE IV
THE SIZE OF THE WAGE SPILLOVERS

	% Δ Affected wage		Spillover share of wage increase
	% Δw	% Δw No spillover	$\frac{\% \Delta w - \% \Delta w_{\text{No spillover}}}{\% \Delta w}$
Overall	0.068*** (0.010)	0.041*** (0.009)	0.397*** (0.119)
Less than high school	0.077*** (0.013)	0.048*** (0.009)	0.370*** (0.078)
Teen	0.081*** (0.015)	0.053*** (0.007)	0.347*** (0.059)
High school or less	0.073*** (0.013)	0.043*** (0.011)	0.402*** (0.100)
Women	0.070*** (0.011)	0.045*** (0.010)	0.359*** (0.120)
Black or Hispanic	0.045*** (0.012)	0.037*** (0.010)	0.179 (0.265)
Tradeable	0.058 (0.073)	0.065** (0.028)	-0.114 (1.157)
Nontradeable	0.056*** (0.014)	0.043*** (0.006)	0.237 (0.191)
Incumbent	0.095*** (0.020)	0.055*** (0.011)	0.422** (0.181)
New entrant	0.019 (0.013)	0.023*** (0.006)	-0.178 (0.748)

6.4 Table IV: the size of wage spillovers (Equation (3) decomposition)

Table IV decomposes the total affected-wage increase into:

- a **direct** component (no-spillover): what wages would do if only sub-minimum jobs are lifted to MW,
- a **spillover** component: the remainder, interpreted as wage increases above MW induced by the policy.

Main messages.

- For the overall workforce, the total affected-wage change is about 6.8%, while the no-spillover benchmark is about 4.1%, so roughly **40% of the wage effect is spillovers**.
- Spillover shares are similar for many high-exposure groups (teens, less education, women), but smaller and imprecisely estimated for Black/Hispanic workers.
- Spillovers are larger for incumbents than for new entrants, consistent with Figure IV: ripple effects reflect wage adjustments for continuing matches.

7 Short synthesis

The paper's event-based bunching approach turns the minimum-wage question into a transparent accounting exercise: minimum wages remove low-wage jobs from below the new floor (missing jobs) and (mostly) relocate them to at/above the floor (excess jobs). Across a long U.S. panel of state hikes, missing jobs are large, excess jobs are similarly large, and net employment effects are close to zero, while wage effects are clearly positive and include nontrivial spillovers. By inspecting the entire wage distribution, the paper also provides a powerful diagnostic: if a specification implies

large employment effects far above the minimum wage, it is likely picking up confounders rather than causal policy effects.

8 Conclusion

8.1 Summary

This paper concludes that U.S. minimum-wage increases primarily **reallocate low-wage jobs into higher-wage bins** rather than destroying them on net. The central empirical object is not aggregate employment, but the **wage distribution of jobs**. When the minimum wage rises, the authors measure (i) *missing jobs below the new minimum wage* and (ii) *excess jobs at and just above the new minimum wage*. Across 138 state minimum-wage hikes from 1979–2016, they find that the missing jobs are large and highly statistically significant, but they are **almost fully offset** by excess jobs above the new minimum wage, so the implied **net employment effect is close to zero**.

Concretely, in the benchmark specification (Table I, col. (1)):

- **Missing jobs below the new minimum wage:** $\Delta b \approx -0.018$ (as a fraction of pre-treatment employment),
- **Excess jobs above the new minimum wage (up to \$5 above):** $\Delta a \approx +0.021$,
- hence **net change in the affected range:** $\Delta e = \Delta a + \Delta b \approx +0.003$ (economically small and statistically insignificant).

At the same time, minimum wages **clearly raise wages** among affected workers: the benchmark estimated increase in affected wages is about **6.8%** (Table I, col. (1)).

The paper also concludes that the wage gains reflect both:

- (i) the **mechanical** effect of lifting sub-minimum wages up to the new floor, and
- (ii) a sizable **spillover/ripple** effect raising wages above the new minimum wage.

The spillover share of the affected wage increase is about **40%** in the benchmark calculation (Table IV: 0.397).

Finally, the paper provides an important **methodological conclusion**: standard two-way fixed effects regressions that use aggregate employment (or broad wage groups) can be misleading because they can be driven by apparent “effects” in **upper-tail wage bins** where minimum wages should not matter. The distribution-based approach is therefore both (i) a more transparent estimator and (ii) a diagnostic tool for confounding.

8.2 Economic intuition: why missing jobs can coexist with near-zero net employment effects

The key intuition is an **accounting-and-reallocation mechanism** inside the wage distribution.

Step 1: binding minimum wages must remove jobs from below the new floor. If the minimum wage becomes binding, jobs paying $w < \text{MW}$ cannot remain at those wages. Thus, the paper expects and indeed finds a large negative Δb (“missing jobs”) below the new minimum wage.

Step 2: the crucial question is where those jobs go. Jobs that disappear from below MW can:

- be **destroyed** (true disemployment), or
- be **reabeled/repriced** (same jobs, higher wages), appearing as **excess employment** at MW and slightly above.

The paper shows the second mechanism dominates: the excess jobs above MW nearly offset the missing jobs below it.

Step 3: spillovers are a natural consequence of wage ladders and internal pay structure.

Firms may raise wages not only to meet the new legal floor but also to maintain wage differentials among adjacent job types or tenure groups. This produces a “ripple” of wage increases above the minimum wage. Empirically, the paper estimates that the no-spillover (mechanical) component of affected wage gains is about 4.1%, while the total is 6.8%, implying a spillover share around 40% (Table IV).

Step 4: why focusing on the wage distribution improves identification. Minimum wages are designed to affect the bottom of the wage distribution. By estimating employment changes *by wage bin* and checking that high-wage bins do not move systematically, the approach limits the scope for confounders (state macro shocks, sectoral changes, etc.) to masquerade as minimum-wage effects.

8.3 What the paper rules out (and what it does not)

The evidence strongly rules out large negative employment effects among low-wage jobs in the U.S. historical data studied. However, the estimates still allow for small positive or small negative net employment changes (confidence intervals around zero). The paper’s main message is therefore:

minimum wages in this setting generate large wage gains and large reallocation within the low-wage distribution, but little net job loss.

8.4 Takeaway

Minimum-wage hikes eliminate jobs paid below the new wage floor but shift almost the same number of jobs into bins at and above the new minimum wage, yielding near-zero net employment effects alongside sizable wage gains (with about 40% coming from spillovers above the floor).